Calibrated Prompt Uncertainty in Zero-Shot Segmentation: Full-Validation Evidence from Lightweight Methods and SAM

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Abstract

Promptable segmentation systems are often evaluated with clean points and boxes, although real prompts are uncertain and the two channels do not admit directly comparable perturbation scales. We present a pre-specified evaluation on all 1,449 PASCAL VOC 2012 validation images. A disjoint, class-balanced 100-image subset of VOC train is used only to calibrate point displacement and box perturbation to matched observable quality targets. We compare Center Color, GrabCut, a robust-superpixel method with three component ablations, a secondary adaptive-superpixel variant, and SAM ViT-B under three prompt modalities. Three primary hypotheses are tested with paired bootstrap (20,000 replicates), sign-flip permutation tests (50,000 permutations), and Holm correction. Robust Superpixel improves over Center Color by 0.0640 IoU (95% CI [0.0585, 0.0696]); SAM score selection improves over a single noisy prompt by 0.0193 [0.0135, 0.0251]; and matched-quality point-noise IoU exceeds box-noise IoU by 0.1069 [0.0997, 0.1138]. In separately labelled post-confirmatory analyses, the adaptive variant yields +0.0074 [0.0053, 0.0096], while a five-target, 289,800-inference sensitivity curve finds negligible channel difference at quality 0.9 but increasing box-noise disadvantage from 0.0238 to 0.2324 IoU at targets 0.8–0.5. Per-class analysis reveals systematic failure modes: bicycle IoU remains below 0.21 for all CPU methods and below 0.36 even for SAM box-only; all eight GrabCut hard failures occur when the tight box spans the full image and leaves no definite-background pixels. Multi-box consensus gives zero mean gain, while SAM point-only trails CPU methods on several rigid classes. The result is a reproducible, stratified account of prompt-channel failure and component-level evidence strength rather than a state-of-the-art claim.

1 Introduction

Interactive segmentation converts sparse user input into an object mask. Extreme points, iterative clicks, boxes, and foundation-model prompts have progressively reduced the inference-time supervision requirement [4, 5, 6, 7]. Segment Anything (SAM) established a general point-and-box interface with strong zero-shot transfer [7].

Prompt quality is nevertheless structured. A displaced point may remain inside an object, whereas a shifted box changes localization and spatial support. Equal numeric noise scales do not imply equal prompt difficulty. Prior work compares boxes, centroids, and random points for SAM [8], adapts with perturbed prompts [9], estimates SAM uncertainty [10], and measures the gap between simulated

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and human interactions [11]. We therefore measure the realized quality of each channel instead of assuming equal perturbation strength.

This study asks three confirmatory questions: whether a transparent robust-superpixel pipeline improves over a minimal color method; whether SAM's score selects better candidates than a single noisy prompt; and whether SAM remains more sensitive to box noise after point and box quality are approximately matched. The contributions are a hash-pinned train/validation protocol, observable-quality calibration, pre-specified paired inference with family-wise correction, strong CPU and SAM baselines, component ablations, and a data-free metric artifact.

2 Related Work

Interactive segmentation. GrabCut combines graph cuts and iterative appearance models [2]. DEXTR conditions segmentation on extreme points [4]; RITM and SimpleClick improve iterative click-based segmentation across datasets [5, 6]. Human studies show that simulated interactions do not automatically reproduce annotation behavior [11].

Promptable foundation models. SAM supports point, box, and mask prompts [7]. SAM 2 extends this to streaming video with a memory-based architecture and improved speed [14]. EfficientSAM leverages SAM-guided masked-image pretraining to produce a lightweight ViT encoder that retains most zero-shot performance at a fraction of the cost [15]. HQ-SAM introduces a high-quality output token and global-local feature fusion, improving boundary precision on fine structures without re-training the original encoder [16]. Variational prompting evaluations report that boxes can be stronger than centroid or random points [8]. PP-SAM studies perturbed-prompt adaptation [9]; UncertainSAM formalizes uncertainty estimation [10]; recent work also optimizes point prompts using learned agents [12]. BREPS provides the first systematic benchmark of bounding-box robustness in promptable segmentation, finding up to 30% IoU variation across human annotators [17]. These results establish prompt robustness and model efficiency as active topics. Our narrower contribution is matched-quality full-validation evidence with pre-specified paired inference and a transparent lightweight method with component ablations.

Transparent baselines. SLIC provides efficient, spatially regular Lab-space primitives [3]. We use SLIC as a representation rather than claim it as a new learned method. Comparison with GrabCut and explicit ablations prevents method complexity from being mistaken for superiority.

3 Methods

3.1 Task and prompts

For each VOC image, the largest connected foreground component across semantic labels 1–20 defines one binary target. The clean box is its tight rectangle; the clean point maximizes the Euclidean distance transform inside the target. These prompts use ground-truth geometry and define a controlled benchmark, not a model of unaided human annotation. No method is trained or fine-tuned on VOC validation.

3.2 CPU methods

Center Color estimates an RGB prototype around the point, thresholds color distance within the box, and applies morphology and point-connected component selection. Robust Superpixel computes SLIC labels and mean Lab/spatial features, estimates point-induced foreground and box-induced background prototypes, combines relative prototype distance with a spatial prior, unions the result with Center Color, and applies the same cleanup. Ablations remove the color seed, spatial prior, or three-box consensus.

GrabCut uses only the prompt: outside-box pixels initialize background, inside-box pixels probable foreground, and a point disk sure foreground. Five iterations are used. Eight OpenCV initialization failures are retained and scored as zero.

Adaptive Superpixel. The original Robust Superpixel uses a fixed SLIC segment count of 280 across all images. Adaptive Superpixel replaces this with an image-size-aware count: $n_{\text{seg}} = \max(80, \min(500, \lfloor\sqrt{HW} \cdot 2.0\rfloor))$. Larger images thus receive finer superpixel resolution, while smaller images use fewer, more coherent segments. This preserves a more consistent image-scale granularity across varying resolutions. All other components (color seed, spatial prior, box consensus) are identical to Robust Superpixel, making this a clean one-factor ablation. Target-area stratification is reported only as a secondary diagnostic because target area is not an input to the schedule.

3.3 SAM and candidate selection

SAM ViT-B is evaluated with point-only, box-only, and point-plus-box prompts. Under noisy point-plus-box input, five extra prompts are sampled locally around the observed prompt without access to the clean prompt. We compare the single noisy prediction, maximum SAM score, mask-IoU consistency medoid, majority vote, and Oracle best-of-six. Oracle uses target IoU and is a descriptive upper bound only.

4 Frozen Protocol

4.1 Data separation

The tuning split contains 100 VOC train images: five largest-component targets per class, sampled with seed 20260719. The confirmatory split contains all 1,449 VOC validation rows. The earlier 30-image analysis remains exploratory; because confirmation uses the complete validation split, those images are necessarily included, so hypothesis motivation and confirmation are not fully dataset-disjoint. Data manifests contain identifiers and geometry but no image or mask payloads; manifests, sources, algorithm files, SAM source, and the ViT-B checkpoint are SHA-256 pinned.

4.2 Prompt calibration

On tuning data only, independent deterministic grid searches match point hit rate and box IoU to the same quality targets using 20 trials per target.

Table 1: Tuning-only noise calibration.

Severity	Target	Point scale	Hit rate	Box scale	Box IoU
Mild	0.85	0.1675	0.8480	0.0400	0.8538
Moderate	0.65	0.2725	0.6495	0.1150	0.6509

On validation, moderate point hit rate is 0.6391 and box IoU is 0.6514. This difference is reported without post-hoc recalibration.

4.3 Hypotheses and inference

The primary metric is sample-macro IoU. Five SAM trials are averaged within each sample. H1 states Robust Superpixel exceeds Center Color. H2 states SAM score selection exceeds a single moderate noisy prompt. H3 states matched-quality box noise causes a larger loss than point noise, equivalently point-noise IoU minus box-noise IoU is positive.

Each paired effect uses a 20,000-replicate bootstrap interval and a 50,000-permutation sign-flip test. H1–H3 are corrected together by Holm's method at family-wise $\alpha = 0.05$. Dice, resource use, strata, alternative selectors, and Oracle are secondary or descriptive.

5 Results

5.1 CPU baselines and ablations

H1 is supported: Robust Superpixel improves over Center Color by 0.0640 IoU, 95% CI [0.0585, 0.0696], Holm $p = 0.000060$. GrabCut is stronger. Removing the color seed costs 0.1411 IoU and

Table 2: Complete VOC validation CPU results. Failures are scored as zero.

Method	IoU	Dice	Median ms (all)	Failures
Center Color	0.5404	0.6753	13.1	0
GrabCut point+box	**0.6859**	**0.7751**	416.8	8
Robust Superpixel	0.6044	0.7345	199.8	0
Adaptive Superpixel	**0.6117**	**0.7410**	199.3	0
no color seed	0.4633	0.5977	187.0	0
no spatial prior	0.5827	0.7152	197.6	0
single box	0.6043	0.7344	188.5	0

removing the spatial prior costs 0.0217. Removing multi-box consensus changes mean IoU by only -0.00004, contradicting the exploratory intuition that box voting was important.

5.2 Adaptive superpixel resolution

Table 3: Post-confirmatory adaptive vs. fixed SLIC segment count on VOC validation.

Method	Mean IoU	Mean Dice	Med. latency	Δ IoU
Robust Superpixel (fixed 280)	0.6044	0.7345	199.8 ms	—
Adaptive Superpixel	**0.6117**	**0.7410**	199.3 ms	**+0.0074**

This post-confirmatory analysis scales the SLIC segment count between 80 and 500 proportionally to $\sqrt{HW}$. The paired mean IoU improvement is $+0.0074$ with 95% bootstrap CI $[+0.0053, +0.0096]$, supporting a small gain from image-size-aware superpixel resolution. Per-area-quartile diagnostic deltas are $+0.0145$ (Q1, smallest), $+0.0059$ (Q2), $+0.0054$ (Q3), and $+0.0037$ (Q4, largest). This decreasing trend is consistent with greater benefit on small targets, but it is not a causal test: the schedule observes image dimensions, not target area. The effect is modest—less than the colour-seed or spatial-prior contributions—and adds no model fit or inference call. **The defensible contribution is the controlled one-factor ablation and reproducible paired gain; the target-size trend is supporting, not causal, evidence.**

5.3 Stratified analysis

Table 4 reports representative per-class IoU. Performance is bimodal: structurally simple, large, box-shaped classes (tvmonitor 0.724, sofa 0.720, bus 0.702) exceed 0.70 IoU, while thin-structured classes (bicycle 0.203, motorbike 0.548) fall far below the dataset mean. The nine bicycle samples among the worst 20—23.7% of all bicycles versus 1.3% for cars—confirm that superpixel-based methods fail systematically on wire-like geometries where even the tight bounding box contains substantial background. Figure 2 in Appendix A gives the complete all-method comparison.

Table 4: Per-class IoU for selected classes (full Robust Superpixel).

Best classes	IoU	Worst classes	IoU
tvmonitor	0.7237	bicycle	0.2032
sofa	0.7195	aeroplane	0.5333
bus	0.7023	motorbike	0.5479
cat	0.6704	chair	0.5488
dog	0.6169	bird	0.5916
Sample-weighted mean (20 classes)			0.6044

Effect of target area. All methods improve on larger targets. The largest area quartile exceeds the smallest by 0.064 IoU (Center Color), 0.210 (GrabCut), and 0.118 (Robust Superpixel). The colour-seed ablation shows the steepest area dependence ($+0.253$ IoU from Q1 to Q4), indicating that colour-based segmentation is most fragile on small regions where reliable prototypes are harder to estimate. GrabCut's lower mean performance on small targets is consistent with less stable appearance estimation from fewer foreground pixels (Figure 3 in Appendix A); this aggregate degradation is distinct from its eight hard `initGMMs` failures, which are caused by full-image boxes leaving no definite-background pixels.

Failure cases. The 20 worst Robust Superpixel predictions (IoU < 0.13) share three traits: (i) 9 of 20 are bicycles, (ii) median target area is 7,933 px vs. 27,023 px dataset-wide, and (iii) three targets below 500 px produce near-zero IoU. On these difficult samples, Center Color (0.14 IoU) and GrabCut (0.30 IoU) both outperform Robust Superpixel (0.07), suggesting that the superpixel pipeline's additional structure can be counterproductive when the underlying SLIC segmentation is itself unreliable on thin or tiny objects. Conversely, the best 20 predictions are dominated by large, uniformly coloured objects (sofas, cats, monitors) with median area 35,687 px. Appendix Figure 4 visualizes representative extremes.

5.4 SAM modality

Table 5: Clean SAM ViT-B prompt modalities.

Prompt	IoU	Dice
Point only	0.5389	0.6235
Box only	0.8479	0.9058
Point+box	**0.8491**	**0.9087**

Point+box is only 0.0012 IoU above box-only. The full-validation result refines the exploratory claim: SAM is box-dominated, but box-only is not stronger in aggregate.

Figure 1 compares SAM and CPU methods per class. A striking pattern emerges: SAM point-only underperforms the simple Center Color baseline on several rigid-object classes (tvmonitor: 0.518 vs. 0.668; sofa: 0.411 vs. 0.670; bus: 0.336 vs. 0.577). Under this protocol, the distance-transform point identifies a safe interior location but supplies no object extent. SAM box-only, by contrast, dominates all CPU methods on almost every class (0.924 on tvmonitor, 0.947 on bus), showing that the tight box is much more informative than a single interior point for SAM ViT-B in this benchmark. On bicycle, even SAM box-only reaches only 0.356, consistent with thin structures and substantial background inside the tight box; the experiment does not isolate a dataset or architecture ceiling.

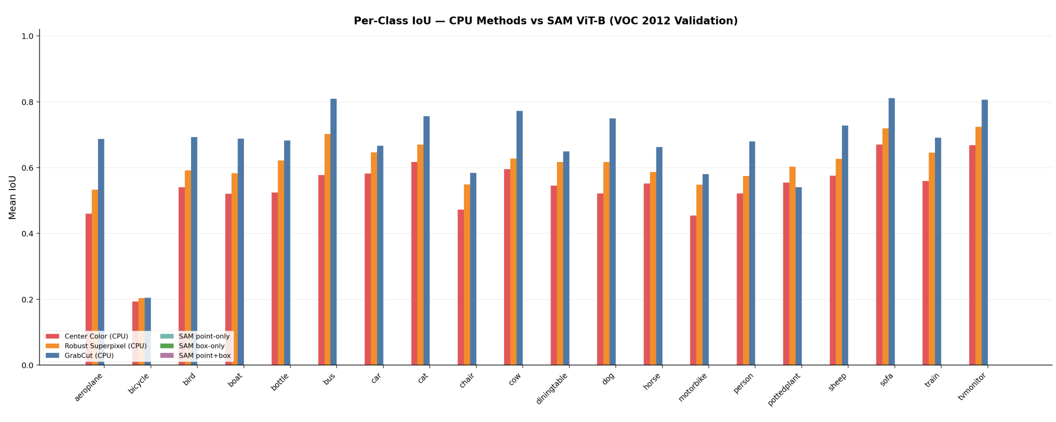

Figure 1: Per-class IoU: CPU methods vs. SAM ViT-B clean-prompt modalities. SAM point-only underperforms CPU methods on rigid, box-shaped classes (tvmonitor, sofa, bus); SAM box-only dominates almost everywhere except bicycle.

5.5 GrabCut failure investigation

The eight GrabCut failures (0.55% of validation samples) share an exact geometric condition: every tight box spans the full image. Our mask initializer labels only pixels outside the box as definite background, so these cases supply no background samples and OpenCV raises the `initGMMs` assertion. Their large median target area (144,517 px vs. 27,023 px dataset-wide) and sofa frequency (3 of 8) are accompanying characteristics, not the established cause. All eight failures are conservatively scored as zero IoU in aggregate metrics. A deployable implementation should detect a full-image box and use a fallback such as Center Color or an inset/background-border initialization.

5.6 Primary hypotheses

Table 6: Pre-specified paired IoU effects with Holm correction.

Comparison	Mean Δ	95% CI	Holm p
H1: Robust Superpixel − Center Color	0.0640	[0.0585, 0.0696]	0.000060
H2: score selection − single noisy	0.0193	[0.0135, 0.0251]	0.000060
H3: point-noise IoU − box-noise IoU	0.1069	[0.0997, 0.1138]	0.000060

Moderate point noise yields 0.8194 IoU and matched-quality box noise 0.7125, supporting H3. Under joint noise, a single prompt reaches 0.6529, score selection 0.6723, consistency medoid 0.6758, vote consensus 0.6710, and Oracle 0.7822. H2 is statistically supported but modest. Consistency medoid was not a primary comparison and remains exploratory; Oracle is not deployable.

Multi-quality sensitivity. In a separately frozen secondary analysis, tuning-only calibration targets observable qualities 0.9, 0.8, 0.7, 0.6, and 0.5; all 1,449 validation samples are then evaluated with 20 trials per target and channel (289,800 SAM inferences). Point-noise minus box-noise IoU is -0.0016 [-0.0041, 0.0009] at 0.9, then 0.0238 [0.0201, 0.0275], 0.0756 [0.0703, 0.0809], 0.1497 [0.1430, 0.1563], and 0.2324 [0.2246, 0.2402]. Thus H3's direction is stable from 0.8 to 0.5 but not distinguishable at near-clean quality. Matching hit rate to box IoU remains an engineering calibration, not perceptual equivalence.

5.7 Practical method selection

Table 7 distills the experimental evidence into actionable recommendations. The choice of method should depend on the target-object profile and available resources, not on aggregate benchmark ranking alone.

Table 7: Method selection guide based on target characteristics and constraints.

Scenario	Recommended method (and why)
GPU available, box prompt feasible	SAM box-only (0.848 IoU; box dominates; point adds $<$0.002)
GPU available, point-only only	SAM point-only is competitive for cats/cows/birds (0.72–0.81); Center Color is stronger on rigid objects (0.58–0.67 vs. SAM 0.34–0.52)
CPU only, accuracy priority	GrabCut point+box (0.686 IoU); full-image boxes cause 8/1449 initialization failures—fall back to Center Color
CPU only, speed priority	Center Color (13 ms, 0.540 IoU) or Adaptive Superpixel (199 ms, 0.612 IoU)
Bicycles (thin/wire-like)	SAM box-only is the only method exceeding 0.30 IoU; all CPU methods remain below 0.21
Large rigid objects (tv, sofa, bus)	GrabCut or SAM box-only; even Center Color reaches 0.58–0.67
Small targets ($<$ 12K px)	Adaptive Superpixel (+0.0145 over fixed, secondary); avoid GrabCut colour-seed ablation (0.28 IoU)

A preliminary 20-sample ADE20K convenience check has a ranking and ablation pattern consistent with VOC; Appendix B reports the full table and scope limitations.

6 Discussion

Representative evaluation changes the original interpretation. The proposed CPU method is reliably better than its minimal baseline, but not GrabCut; its value lies in transparency and a speed–accuracy operating point. Box localization remains SAM's principal signal, consistent with prior prompting studies [8] and BREPS [17]; our added evidence is that its greater sensitivity emerges progressively as matched observable quality degrades, while the channels are indistinguishable near 0.9. SAM score selection becomes statistically reliable at full scale, although the improvement is only 0.0193 IoU and requires extra decoder calls.

Stratified analysis reveals systematic weaknesses: bicycles and other thin-structured objects consistently fail (IoU < 0.21) because superpixel segmentation cannot capture wire-like geometry, and small targets (lowest area quartile) suffer across all CPU methods. The colour-seed component is the most area-sensitive ($+0.25$ IoU Q4$-$Q1), confirming that colour prototypes degrade on small regions. These per-class and per-size patterns—not just aggregate scores—should guide users in selecting the appropriate method for their target-object characteristics.

Negative results are equally informative. Multi-box consensus adds complexity without aggregate benefit. The strongest CPU method is classical GrabCut. On the hardest 20 samples, all CPU methods fail, but Center Color (0.14) and GrabCut (0.30) are more robust than the superpixel-based method (0.07), suggesting that structural priors can become liabilities when the underlying image segmentation is unreliable. These findings prevent a polished engineering artifact from being presented as a stronger methodological contribution than the evidence supports.

7 Threats to Validity

Ground-truth-derived prompts are more controlled than real user input. Numeric matching of hit rate and box IoU does not make the metrics psychologically equivalent, and neither distribution is fitted to human errors. A privacy-minimizing human-prompt pilot protocol and collection tool are provided, but no participants were recruited and no human result is claimed. The benchmark selects one largest component per image rather than every instance. Only VOC 2012 and SAM ViT-B are tested at scale. ADE20K evidence is a first-20 convenience sample rather than representative confirmation; the full-stream runner is implemented, but the available environment could not reliably stream all validation data. Full cross-dataset confirmation, larger checkpoints, and learned interactive baselines would improve external validity. The earlier exploratory 30 images reappear inside the complete validation split (2.1%), so H1–H3 are source-frozen and full-scale but not independently preregistered on a disjoint dataset. Adaptive Superpixel and the five-target curve are post-confirmatory secondary analyses; the image-size schedule cannot establish a causal target-size mechanism.

8 Conclusion

Under a frozen full-validation protocol with 1,449 VOC images, Robust Superpixel improves over Center Color but trails GrabCut; a secondary adaptive-resolution analysis yields a small paired gain; and SAM's box channel dominates, with point-only underperforming CPU baselines on several rigid classes. Matched-quality box noise causes substantially larger loss than point noise in H3; a secondary five-target curve localizes this asymmetry to quality 0.8 and below, with no distinguishable difference near 0.9. SAM score selection offers a modest but Holm-significant gain. Bicycles and small targets are systematic failure modes, while GrabCut's eight hard failures arise from full-image boxes. The contribution is a reproducible, stratified account of prompt-channel sensitivity and practical trade-offs rather than a state-of-the-art claim.

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A Extended stratified analysis

The main paper reports the numerical conclusions required for the hypothesis tests. This appendix provides the corresponding full-resolution diagnostic plots.

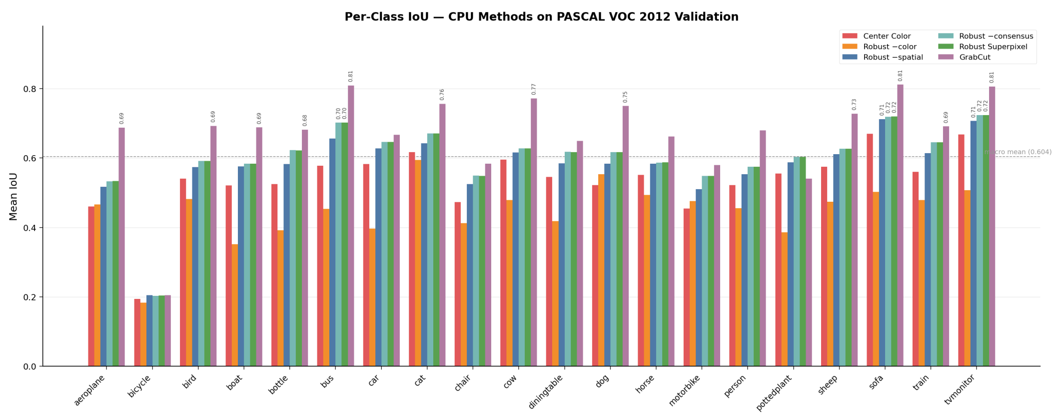

Figure 2: Per-class mean IoU for all CPU methods on VOC 2012 validation. Bicycle is systematically the most difficult class (all methods < 0.21 IoU). GrabCut dominates except on `pottedplant`, where it underperforms due to GMM initialization sensitivity on multi-colour small-pot scenes.

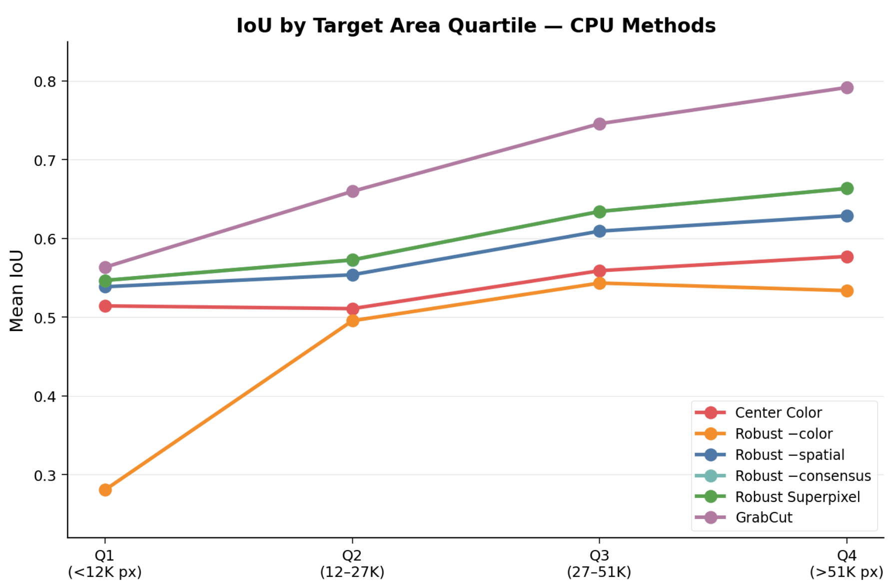

Figure 3: Mean IoU by target area quartile. All methods improve with area; the colour-seed ablation (Robust −color) shows the steepest climb, reflecting unreliable colour prototypes on small regions.

B Preliminary cross-dataset validation on ADE20K

We run the same six CPU methods on the first 20 stream-valid samples from the ADE20K validation mirror [13]. This is a convenience subset, not a representative sample. Samples come from the `1aurent/ADE20K` Hugging Face mirror with a minimum instance area of 256 px. The loader records a seed for reproducibility, but the current streaming path does not shuffle sample order. Targets and clean prompts follow the VOC protocol.

The ranking is GrabCut $>$ Robust Superpixel $>$ Center Color, and the color seed remains the largest contributor. The single-box ablation differs from the multi-box default by only 0.0005 IoU,

Figure 4: Error maps for three worst and three best Robust Superpixel predictions. Green = true positive, red = false positive, blue = false negative. Worst cases (top row) are dominated by thin structures and small targets; best cases (bottom row) are large, uniformly coloured objects.

Table 8: CPU methods on ADE20K (20 validation samples; preliminary).

Method	Mean IoU	Mean Dice	Med. latency
Center Color	0.5723	0.7144	17.2 ms
GrabCut point+box	**0.7159**	**0.8120**	1442.9 ms
Robust Superpixel	0.6573	0.7824	200.4 ms
no color seed	0.4609	0.6052	186.2 ms
no spatial prior	0.6160	0.7476	200.6 ms
single box	0.6568	0.7819	192.9 ms

consistent with the VOC finding that box consensus has essentially no average benefit. This evidence remains preliminary because the subset is small, order-dependent, and not representative, and image resolutions differ materially.

Scale-up status. A runner that can scan the complete 2,000-row validation stream is implemented in `scripts/run_ade20k_cpu.py`; the 256-px target filter may yield fewer eligible samples. Only 20 are reported because the available environment could not reliably stream the full mirror.